Real Analysis

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1 Preliminaries

Definition 1.1. *(Limit Superior and Limit Inferior)*
Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets. The limit superior and the limit inferior are

$$\limsup A_n = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n, \quad \liminf A_n = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n.$$

Remark.
A point x lies in $\limsup A_n$ if and only if x belongs to infinitely many of the sets A_n. It lies in $\liminf A_n$ if and only if x fails to belong to only finitely many of the sets A_n.

Definition 1.2. *(Sigma-Algebra)*
A family $\mathcal{A} \subset 2^X$ is a sigma-algebra on X if
- $\varnothing \in \mathcal{A}$;
- $A \in \mathcal{A}$ implies $A^c \in \mathcal{A}$;
- $\{A_n\}_{n=1}^{\infty} \subset \mathcal{A}$ implies $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Definition 1.3. *(Borel Sigma-Algebra)*
The Borel sigma-algebra $\mathcal{B}$ on $\mathbb{R}$ is the smallest sigma-algebra containing every open subset of $\mathbb{R}$. A set in $\mathcal{B}$ is a Borel set.

Remark.
An F_σ set is a countable union of closed sets. A G_δ set is a countable intersection of open sets. Both are Borel.

Lemma 1.1.
Every open set $U \subset \mathbb{R}$ is an at most countable union of pairwise disjoint open intervals.

Theorem 1.1. *(Cantor)*
The interval $[0, 1]$ is uncountable.

Proof.
Suppose $[0, 1] = \{x_1, x_2, \ldots\}$. Trisect $I_0 = [0, 1]$. At least one of the two closed outer thirds misses x_1; call it I_1. Trisect I_1 and choose a closed third I_2 that misses x_2.

Continuing gives a nested sequence $I_0 \supset I_1 \supset \cdots$ of nonempty closed intervals with $x_n \notin I_n$. Their intersection contains a point x. Then $x \in [0,1]$ and x equals no x_n. ■

2 Lebesgue Measure

2.1 Outer Measure

Definition 2.1. *(Outer Measure)*
For $A \subset \mathbb{R}$, the outer measure of A is

$$m^*(A) = \inf\left\{\sum_{k=1}^{\infty} l(I_k) : A \subset \bigcup_{k=1}^{\infty} I_k\right\},$$

where each I_k is a bounded open interval and $l(I_k)$ is its length. The empty set admits the empty cover, so $m^*(\varnothing) = 0$.

Remark.
- $m^*(A) \geq 0$, and $m^*(\varnothing) = 0$. A set with $m^*(A) = 0$ is a null set.
- If $A \subset B$, then $m^*(A) \leq m^*(B)$.

Proposition 2.1.
Every countable set $A \subset \mathbb{R}$ satisfies $m^*(A) = 0$.

Proposition 2.2.
If $I \subset \mathbb{R}$ is an interval, then $m^*(I) = l(I)$, where an unbounded interval has length ∞.

Proof.
If $I = [a,b]$, then $I \subset (a-\varepsilon, b+\varepsilon)$, so $m^*(I) \leq l(I)$. Conversely, any open cover of I has a finite subcover, and the lengths of those intervals sum to at least $l(I)$. Thus $m^*(I) \geq l(I)$.
A bounded interval is pinched between closed intervals whose lengths approach $l(I)$. An unbounded interval contains closed subintervals of arbitrary finite length, so its outer measure is infinite. ■

Proposition 2.3.
For every $A \subset \mathbb{R}$ and every $x \in \mathbb{R}$, $m^*(A + x) = m^*(A)$.

Proposition 2.4. *(Countable Subadditivity)*

$$m^*\left(\bigcup_{k=1}^{\infty} E_k\right) \leq \sum_{k=1}^{\infty} m^*(E_k).$$

Proof.
The claim is immediate if some $m^*(E_k)$ is infinite. Otherwise, fix $\varepsilon > 0$. Cover E_k by open intervals whose lengths sum to less than $m^*(E_k) + \varepsilon/2^k$. These intervals together cover the union, and their lengths sum to less than $\sum_k m^*(E_k) + \varepsilon$. Let $\varepsilon \to 0$. ■

Example.
Let C be the middle-thirds Cantor set, and let C_n be the union of the 2^n closed intervals of length 3^{-n} remaining at stage n. Then $C \subset C_n$, so

$$m^*(C) \leq m^*(C_n) \leq \left(\frac{2}{3}\right)^n \to 0.$$

Thus $m^*(C) = 0$.

Proposition 2.5.
If $d(E_1, E_2) > 0$, then $m^*(E_1 \cup E_2) = m^*(E_1) + m^*(E_2)$.

Proof.
Subadditivity gives one inequality. For the other, assume both outer measures are finite and set $\delta = d(E_1, E_2)$ and $E = E_1 \cup E_2$. Given $\varepsilon > 0$, cover E by open intervals of total length less than $m^*(E) + \varepsilon$, and subdivide them into open intervals of length less than $\delta/2$. No such interval can meet both E_1 and E_2. Enlarging the pieces by a factor $\lambda > 1$ if needed to keep them open and covering, one gets

$$m^*(E_1) + m^*(E_2) < \lambda(m^*(E) + \varepsilon).$$

Let $\lambda \to 1$ and $\varepsilon \to 0$. ■

2.2 Measurable Sets

Definition 2.2. *(Carathéodory)*
A set $E \subset \mathbb{R}$ is Lebesgue measurable if, for every $A \subset \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c).$$

Remark.
Countable subadditivity always yields $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$. Measurability is the reverse inequality.

Proposition 2.6.
If $m^*(E) = 0$, then E is measurable.

Proof.
For every A, $m^*(A \cap E) \leq m^*(E) = 0$, so

$$m^*(A) \geq m^*(A \cap E^c) = m^*(A \cap E) + m^*(A \cap E^c).$$

■

Proposition 2.7.
If E_1 and E_2 are measurable, then so is $E_1 \cup E_2$. By De Morgan, $E_1 \cap E_2$ and $E_1 \setminus E_2$ are measurable as well.

Proof.
Let $A \subset \mathbb{R}$. Measurability of E_1 and then of E_2 gives

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c). \end{aligned}$$

The first two sets on the right are disjoint and their union is $A \cap (E_1 \cup E_2)$, while the third is $A \cap (E_1 \cup E_2)^c$. Subadditivity closes the estimate. ■

Corollary 2.1.
- If E_1 and E_2 are measurable, then $E_1 \cap E_2$ is measurable.
- A finite intersection of measurable sets is measurable.

Proposition 2.8. *(Finite Additivity)*
Let $\{E_k\}_{k=1}^n$ be pairwise disjoint measurable sets. Then for every $A \subset \mathbb{R}$,

$$m^*\left(A \cap \bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} m^*(A \cap E_k).$$

Proof.
The case $n = 1$ is immediate. If the identity holds up through $n - 1$, measurability of E_n gives

$$m^*\left(A \cap \bigcup_{k=1}^{n} E_k\right) = m^*(A \cap E_n) + m^*\left(A \cap \bigcup_{k=1}^{n-1} E_k\right),$$

and the inductive hypothesis finishes the proof. ■

Proposition 2.9.
A countable union of measurable sets is measurable.

Proof.
Let E_i be measurable and set $F_1 = E_1$, $F_j = E_j \setminus \bigcup_{i<j} E_i$. The sets F_j are measurable and pairwise disjoint, and $E = \bigcup_j E_j = \bigcup_j F_j$. The partial unions $G_n = \bigcup_{j=1}^{n} F_j$ are measurable and increase to E. For every A,

$$m^*(A) = m^*(A \cap G_n) + m^*(A \cap G_n^c) \geq \sum_{j=1}^{n} m^*\big(A \cap F_j\big) + m^*(A \cap E^c).$$

Let $n \to \infty$. Subadditivity of the series $\sum_j m^*\big(A \cap F_j\big)$ yields the Carathéodory inequality for E. ■

Proposition 2.10. *(Countable Additivity)*
Let $\{E_i\}_{i=1}^{\infty}$ be pairwise disjoint measurable sets. Then for every $A \subset \mathbb{R}$,

$$m^*\left(A \cap \bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m^*(A \cap E_i).$$

Proof.
Finite additivity gives

$$\sum_{i=1}^{n} m^*(A \cap E_i) = m^*\left(A \cap \bigcup_{i=1}^{n} E_i\right) \leq m^*\left(A \cap \bigcup_{i=1}^{\infty} E_i\right).$$

Let $n \to \infty$. Countable subadditivity supplies the reverse inequality. ■

Remark.
The measurable sets form a sigma-algebra, the Lebesgue sigma-algebra, written $\mathcal{M}$.

Proposition 2.11.
Let $E \subset A$ be measurable and $m^*(E) < \infty$. If A is measurable, then

$$m^*(A \setminus E) = m^*(A) - m^*(E).$$

Proof.
Measurability of E gives $m^*(A) = m^*(E) + m^*(A \setminus E)$. ■

Proposition 2.12.
Every interval is measurable.

Proof.
It is enough to treat $E = (a, +\infty)$. Let $A \subset \mathbb{R}$. If $a \notin A$, the two pieces below are $A \cap (-\infty, a)$ and $A \cap (a, +\infty)$; the point a does not affect outer measure. Cover A by open intervals I_k, and split each I_k into $I_k \cap E^c$ and $I_k \cap E$. Then

$$m^*(A \cap E^c) + m^*(A \cap E) \leq \sum_k l(I_k).$$

Taking the infimum over covers gives the Carathéodory inequality. ■

Theorem 2.1.
For $E \subset \mathbb{R}$, the following are equivalent.

- E is measurable.
- For every $\varepsilon > 0$ there is an open set $U \supset E$ with $m^*(U \setminus E) < \varepsilon$.
- There is a G_δ set $G \supset E$ with $m^*(G \setminus E) = 0$.
- For every $\varepsilon > 0$ there is a closed set $F \subset E$ with $m^*(E \setminus F) < \varepsilon$.
- There is an F_σ set $F \subset E$ with $m^*(E \setminus F) = 0$.

Proof.
Assume E is measurable. If $m^*(E) < \infty$, choose open intervals covering E whose lengths sum to less than $m^*(E) + \varepsilon$, and let U be their union. Subtraction gives $m^*(U \setminus E) < \varepsilon$. If $m^*(E) = \infty$, write E as a countable union of disjoint measurable sets of finite measure and take a small open neighborhood of each.
Given such neighborhoods U_n with $m^*(U_n \setminus E) < 1/n$, the intersection $G = \bigcap_n U_n$ is a G_δ set containing E and $m^*(G \setminus E) = 0$. Then $G \setminus E$ is a null set, hence measurable, and $E = G \setminus (G \setminus E)$ is measurable.
The closed and F_σ conditions are the same statements applied to E^c. ■

Remark.

- For every $E \subset \mathbb{R}$ there is a G_δ set $G \supset E$ with $m^*(G) = m^*(E)$. Such a G is a measurable cover of E.

- A set E is measurable if and only if there is an F_σ set $F \subset E$ with $m^*(F) = m^*(E)$.

Definition 2.3. *(Lebesgue Measure)*
Lebesgue measure is the restriction of outer measure to the Lebesgue sigma-algebra. It is written $m(E)$.

Definition 2.4. *(Symmetric Difference)*
For $A, B \subset \mathbb{R}$, the symmetric difference is $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

Theorem 2.2.
Let E be measurable with $m(E) < \infty$. For every $\varepsilon > 0$ there exist finitely many pairwise disjoint open intervals $I_1, \cdots, I_n$ such that

$$m\left(E \Delta \bigcup_{k=1}^{n} I_k\right) < \varepsilon.$$

Proof.
Choose an open set $V \supset E$ with $m(V \setminus E) < \varepsilon/2$. Write V as a countable union of pairwise disjoint open intervals I_k. Finite measure of E forces $m(V) < \infty$, so the tail $\sum_{k>n} l(I_k)$ can be made smaller than $\varepsilon/2$. The symmetric difference of E with the finite union is covered by $V \setminus E$ and that tail. ■

2.3 Continuity of Measure

Definition 2.5. *(Monotone Sequences of Sets)*
A sequence $\{E_i\}$ is increasing if $E_i \subset E_{i+1}$, and decreasing if $E_i \supset E_{i+1}$.

Proposition 2.13.
Let $\{E_i\}$ be measurable.
- If $\{E_i\}$ is increasing, then $m\left(\bigcup_{i=1}^{\infty} E_i\right) = \lim_{i\to\infty} m(E_i)$.
- If $\{E_i\}$ is decreasing and $m(E_1) < \infty$, then $m\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i\to\infty} m(E_i)$.

Proof.
For the increasing case, set $E_0 = \varnothing$ and $F_i = E_i \setminus E_{i-1}$. The sets F_i are measurable and pairwise disjoint, and $\cup_i F_i = \cup_i E_i$. Countable additivity gives

$$m\left(\bigcup_i E_i\right) = \sum_{i=1}^{\infty}(m(E_i) - m(E_{i-1})) = \lim_{i\to\infty} m(E_i).$$

The decreasing case follows by applying the increasing case to $E_1 \setminus E_i$. ■

Corollary 2.2.
If $\{E_i\}$ is measurable, then $m(\liminf E_i) \le \liminf m(E_i)$.

Proof.
The sets $F_k = \bigcap_{i\ge k} E_i$ are measurable and increase to $\liminf E_i$, and $m(F_k) \le m(E_k)$. Therefore

$$m(\liminf E_i) = \lim_{k\to\infty} m(F_k) \le \liminf_{k\to\infty} m(E_k).$$

■

Definition 2.6. *(Almost Everywhere)*
A property holds almost everywhere on a measurable set E, written a.e., if the set of points in E where it fails is a null set.

Lemma 2.1. *(Borel--Cantelli)*
Let $\{E_i\}$ be measurable and $\sum_{i=1}^{\infty} m(E_i) < \infty$. Then almost every $x \in \mathbb{R}$ belongs to only finitely many of the sets E_i.

Proof.
The sets $G_k = \bigcup_{i\ge k} E_i$ decrease, and $m(G_k) \le \sum_{i\ge k} m(E_i) < \infty$. Hence

$$m(\limsup E_i) = m\left(\bigcap_{k=1}^{\infty} G_k\right) = \lim_{k\to\infty} m(G_k) = 0.$$

■

Lemma 2.2.
Let $E \subset \mathbb{R}$ be bounded and measurable. If there is an uncountable set Λ such that the translates $\{E + \lambda\}_{\lambda\in\Lambda}$ are pairwise disjoint, then $m(E) = 0$.

2.4 Nonmeasurable Sets

Definition 2.7. *(Rational Equivalence)*
For $x, y \in \mathbb{R}$, write $x \sim y$ when $y - x \in \mathbb{Q}$. This is an equivalence relation.

Remark.
Let $E \subset \mathbb{R}$. By the axiom of choice, the quotient $E/\sim$ has a set $C_E \subset E$ of representatives, and

- $d - c \notin \mathbb{Q}$ whenever $c, d \in C_E$ are distinct, so the translates $\{C_E + \lambda\}_{\lambda \in \mathbb{Q}}$ are pairwise disjoint;
- for every $x \in E$ there exist $c \in C_E$ and $q \in \mathbb{Q}$ with $x = c + q$.

Theorem 2.3. *(Vitali)*
Every $E \subset \mathbb{R}$ with $m^*(E) \neq 0$ contains a nonmeasurable set.

Proof.
Passing to $E \cap [-n, n]$ for large n, one may assume E is bounded and $m^*(E) > 0$. Let $V = C_E$. The rational translates of V are pairwise disjoint, and $E \subset \bigcup_{q \in \mathbb{Q}} (V + q)$. If V were measurable and $m(V) = 0$, then $m^*(E) = 0$. If $m(V) > 0$, the countably many disjoint translates $V + q$ with $q \in \mathbb{Q} \cap [-1, 1]$ would all sit in a fixed bounded interval and the sum of their measures would be infinite. Thus V is not measurable. ■

Corollary 2.3.
There exist disjoint sets $A, B \subset \mathbb{R}$ with $m^*(A \cup B) < m^*(A) + m^*(B)$.

Proof.
If outer measure were additive on every pair of disjoint sets, then every set would satisfy the Carathéodory identity and would be measurable, contradicting Vitali's theorem. ■

Definition 2.8. *(Middle-Thirds Cantor Set)*
The middle-thirds Cantor set is

$$C = \left\{ \sum_{n=1}^{\infty} \frac{a_n}{3^n} : a_n \in \{0, 2\} \right\}.$$

Proposition 2.14. *(Cantor--Lebesgue Function)*
On C, the map

$$\varphi\left(\sum_{n=1}^{\infty} \frac{a_n}{3^n} \right) = \sum_{n=1}^{\infty} \frac{b_n}{2^n}, \quad b_n = \frac{a_n}{2},$$

extends to a continuous nondecreasing surjection $\varphi : [0,1] \to [0,1]$, constant on each open interval in the complement of C.

Proposition 2.15.
The function $\psi : [0,1] \to [0,1]$ given by $\psi(x) = \frac{1}{2}(\varphi(x) + x)$ is strictly increasing, and $m(\psi(C)) = \frac{1}{2}$.

Proof.
Since φ is nondecreasing and $x \mapsto x$ is strictly increasing, ψ is strictly increasing. It is continuous, $\psi(0) = 0$, and $\psi(1) = 1$, so ψ maps $[0,1]$ onto $[0,1]$. Let $D = [0,1] \setminus C = \bigcup_k I_k$, where the I_k are the pairwise disjoint open intervals removed in forming C and $\sum_k l(I_k) = 1$. On each I_k the function φ is constant, so $l(\psi(I_k)) = l(I_k)/2$. These image intervals are pairwise disjoint and

$$\sum_{k=1}^{\infty} l(\psi(I_k)) = \frac{1}{2}.$$

Therefore $m(\psi(C)) = 1 - \frac{1}{2} = \frac{1}{2}$. ■

Corollary 2.4.
There exists a Lebesgue measurable set that is not Borel.

Proof.
Since $m(\psi(C)) = \frac{1}{2} > 0$, Vitali's theorem supplies a nonmeasurable set $E \subset \psi(C)$. The preimage $\psi^{-1}(E)$ is contained in C, so it is a null set and hence Lebesgue measurable. It is not Borel: ψ is a homeomorphism of $[0,1]$ onto itself, and the image of a Borel set under a homeomorphism is Borel, but $E = \psi(\psi^{-1}(E))$ is not Borel. ■

3 Measurable Functions

3.1 Measurable Functions

Proposition 3.1.
Let E be measurable and $f : E \to \overline{\mathbb{R}}$, where $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$. The following are equivalent.

- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) > c\}$ is measurable.
- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) \geq c\}$ is measurable.
- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) < c\}$ is measurable.
- For every $c \in \mathbb{R}$, the set $\{x \in E : f(x) \leq c\}$ is measurable.

Proof.
The identities

$$\{f \geq c\} = \bigcap_{k=1}^{\infty} \left\{ f > c - \frac{1}{k} \right\}, \quad \{f > c\} = \bigcup_{k=1}^{\infty} \left\{ f \geq c + \frac{1}{k} \right\}$$

pass between the first two conditions. Complements give the other two. ■

Definition 3.1. *(Lebesgue Measurable Function)*
Let $f : E \to \overline{\mathbb{R}}$. If E is measurable and one of the four conditions above holds, then f is a Lebesgue measurable function.

Remark.
If f is measurable, then $\{x \in E : f(x) = c\}$ is measurable for every $c \in \overline{\mathbb{R}}$. The converse is false.

Proposition 3.2.
A function $f : E \to \mathbb{R}$ is measurable if and only if $f^{-1}(U)$ is measurable for every open $U \subset \mathbb{R}$.

Corollary 3.1.
- Every continuous function is measurable.
- If $f : E \to \mathbb{R}$ is measurable, $g : F \to \mathbb{R}$ is continuous, and $f(E) \subset F$, then $g \circ f$ is measurable.

Proposition 3.3.
- If f is measurable and $g = f$ almost everywhere, then g is measurable.
- Let $E_0 \subset E$ be measurable. Then f is measurable on E if and only if its restrictions to E_0 and to $E \setminus E_0$ are measurable.

Proposition 3.4.
Let f and g be measurable and finite almost everywhere. Then $\alpha f + \beta g$ and fg are measurable for all $\alpha, \beta \in \mathbb{R}$.

Proof.
For every $c \in \mathbb{R}$,

$$\{x \in E : f(x) + g(x) < c\} = \bigcup_{q \in \mathbb{Q}} (\{g < c - q\} \cap \{f < q\}),$$

which is measurable. The square f^2 is measurable because $\{f^2 > c\} = \{f > \sqrt{c}\} \cup \{f \leftarrow \sqrt{c}\}$ for $c \geq 0$. The product is measurable because

$$fg = \frac{(f+g)^2 - (f-g)^2}{4}.$$

■

Proposition 3.5.
If $f_1, \cdots, f_n$ are measurable, then so are $\max\{f_1, \cdots, f_n\}$ and $\min\{f_1, \cdots, f_n\}$.

Definition 3.2. *(Positive and Negative Parts)*

$$f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\}.$$

Remark.
- $f = f^+ - f^-$ and $|f| = f^+ + f^-$.
- f is measurable if and only if both f^+ and f^- are measurable.

Proposition 3.6.
If $\{f_i\}_{i=1}^{\infty}$ are measurable, then $\sup_i f_i$, $\inf_i f_i$, $\limsup f_i$, and $\liminf f_i$ are measurable.

Corollary 3.2.
An almost-everywhere pointwise limit of measurable functions is measurable.

Definition 3.3. *(Characteristic Function)*
For $A \subset \mathbb{R}$, the characteristic function of A is

$$\chi_A(x) = \begin{cases} 1 \text{ if } x \in A \\ 0 \text{ if } x \notin A \end{cases}.$$

Remark.
The function χ_A is measurable if and only if A is measurable.

Definition 3.4. *(Simple Function)*
A measurable function $f : E \to \mathbb{R}$ is simple if its range is finite. If the distinct values are $c_1, \cdots, c_n$ and $E_k = \{x \in E : f(x) = c_k\}$, then

$$f = \sum_{k=1}^{n} c_k \chi_{E_k}.$$

3.2 Approximation by Simple Functions

Lemma 3.1.
Let $f : E \to \mathbb{R}$ be bounded and measurable. For every $\varepsilon > 0$ there exist simple functions φ_ε and ψ_ε on E with $\varphi_\varepsilon \le f \le \psi_\varepsilon$ and $0 \le \psi_\varepsilon - \varphi_\varepsilon < \varepsilon$.

Proof.
Suppose $f(E) \subset (c, d)$. Choose $c = y_0 < \cdots < y_n = d$ with $y_k - y_{k-1} < \varepsilon$, and set $E_k = f^{-1}([y_{k-1}, y_k))$. The functions $\varphi_\varepsilon = \sum_k y_{k-1}\chi_{E_k}$ and $\psi_\varepsilon = \sum_k y_k \chi_{E_k}$ have the required properties. ■

Theorem 3.1.
A function $f : E \to \overline{\mathbb{R}}$ is measurable if and only if there is a sequence of simple functions $\{\varphi_n\}$ with $\varphi_n \to f$ at every point of E and $|\varphi_n| \le |f|$. If $f \ge 0$, the sequence may be chosen increasing.

Proof.
A pointwise limit of simple functions is measurable. Conversely, since $f = f^+ - f^-$ and $|f| = f^+ + f^-$, it is enough to treat $f \ge 0$. Let $E_n = \{x \in E : f(x) \le n\}$. On E_n the previous lemma supplies a simple function lying between 0 and f and within $1/n$ of f. Extend it by the value n on $E \setminus E_n$, and call the result φ_n. Then $0 \le \varphi_n \le f$. If $f(x)$ is finite and $n \ge f(x)$, then $0 \le f(x) - \varphi_n(x) < 1/n$. If $f(x) = +\infty$, then $\varphi_n(x) = n \to +\infty$. The running maxima $\hat{\varphi}_n = \max\{\varphi_1, \cdots, \varphi_n\}$ are simple, increase, and converge to f. ■

Theorem 3.2. *(Egorov)*
Let E be measurable with $m(E) < \infty$, and let $f_n : E \to \mathbb{R}$ be measurable. If f_n converges pointwise to f, then for every $\varepsilon > 0$ there is a closed set $F \subset E$ with $m(E \setminus F) < \varepsilon$ such that f_n converges uniformly on F.

Remark.
- Pointwise convergence may be weakened to almost-everywhere convergence.
- If $m(E) = \infty$ and $f_n \to f$ almost everywhere, then for every $M > 0$ there is a closed set $F_M \subset E$ with $m(F_M) > M$ on which the convergence is uniform.

Theorem 3.3. *(Lusin)*
Let $f : E \to \mathbb{R}$ be measurable. For every $\varepsilon > 0$ there exist a closed set $F \subset E$ and a continuous function $g : \mathbb{R} \to \mathbb{R}$ such that $m(E \setminus F) < \varepsilon$ and g agrees with f on F.

4 Lebesgue Integral

4.1 Bounded Functions

Definition 4.1. *(Integral of a Simple Function)*
If $\varphi = \sum_{k=1}^{n} c_k \chi_{E_k}$ is simple, with the sets E_k pairwise disjoint and measurable, its Lebesgue integral is

$$\int_E \varphi = \sum_{k=1}^{n} c_k m(E_k).$$

Remark.
The integral of simple functions is linear, additive over measurable subsets, monotone, and satisfies the triangle inequality.

Definition 4.2. *(Upper and Lower Integrals)*
Let E be measurable with $m(E) < \infty$, and let $f : E \to \mathbb{R}$ be bounded. The lower and upper Lebesgue integrals are

$$\underline{\int_E} f = \sup\left\{ \int_E \varphi : \varphi \text{ simple}, \varphi \le f \right\}, \quad \overline{\int_E} f = \inf\left\{ \int_E \psi : \psi \text{ simple}, \psi \ge f \right\}.$$

If they agree, f is Lebesgue integrable and the common value is written $\int_E f$.

Theorem 4.1.
Let $f : [a, b] \to \mathbb{R}$ be bounded. If f is Riemann integrable, then f is Lebesgue integrable and the two integrals agree.

Proof.
Every lower Darboux sum is the integral of a simple function lying below f, and every upper Darboux sum is the integral of a simple function lying above f. Therefore

$$\underline{\int_a^b} f\,\mathrm{d}x \le \underline{\int_{[a,b]}} f \le \overline{\int_{[a,b]}} f \le \overline{\int_a^b} f\,\mathrm{d}x.$$

Riemann integrability forces equality throughout. ■

Theorem 4.2.
Let E be measurable with $m(E) < \infty$. Every bounded measurable $f : E \to \mathbb{R}$ is Lebesgue integrable.

Theorem 4.3. *(Uniform Convergence)*
Let E be measurable with $m(E) < \infty$, and let $\{f_n\}$ be measurable and uniformly bounded. If f_n converges uniformly to f, then f is integrable and $\int_E f = \lim_{n\to\infty} \int_E f_n$.

Remark.
Pointwise convergence is not enough. The tents

$$f_n(x) = n(1 - nx)\chi_{(0,\frac{1}{n})}(x)$$

converge pointwise to 0, while $\int_0^1 f_n = \frac{1}{2}$ for every n.

Theorem 4.4. *(Bounded Convergence)*
Let E be measurable with $m(E) < \infty$, and let $\{f_n\}$ be measurable and uniformly bounded. If f_n converges pointwise to f, then f is integrable and

$$\int_E f = \lim_{n\to\infty} \int_E f_n.$$

4.2 Nonnegative Functions

Definition 4.3. *(Integral of a Nonnegative Function)*
Let $f : E \to \overline{\mathbb{R}}$ be nonnegative and measurable. Define

$$\int_E f = \sup\left\{ \int_E h : h \text{ bounded and measurable}, 0 \le h \le f, m(\{h \neq 0\}) < \infty \right\}.$$

If $\int_E f < \infty$, then f is Lebesgue integrable.

Theorem 4.5. *(Chebyshev)*
Let $f : E \to \overline{\mathbb{R}}$ be nonnegative and measurable. For every $\lambda > 0$,

$$m(\{x \in E : f(x) \geq \lambda\}) \leq \left(\frac{1}{\lambda}\right) \int_E f.$$

Proof.
Let $E_\lambda = \{f \geq \lambda\}$. If $m(E_\lambda) < \infty$, the function $h = \lambda \chi_{E_\lambda}$ is an admissible competitor, so $\lambda m(E_\lambda) = \int_E h \leq \int_E f$. If $m(E_\lambda) = \infty$, the same estimate on $E_\lambda \cap [-n, n]$ sends the left side to infinity, and the integral is infinite as well. ■

Proposition 4.1.
Let $f : E \to \overline{\mathbb{R}}$ be nonnegative and measurable. Then $\int_E f = 0$ if and only if $f = 0$ almost everywhere.

Proof.
If the integral vanishes, Chebyshev gives $m\left(\left\{f \geq \frac{1}{n}\right\}\right) \leq n \int_E f = 0$, so $\{f > 0\} = \cup_n \left\{f \geq \frac{1}{n}\right\}$ is a null set. Conversely, if $f = 0$ almost everywhere and $0 \leq h \leq f$ is bounded with support of finite measure, then $h = 0$ almost everywhere, so $\int_E h = 0$.
■

Lemma 4.1. *(Fatou)*
Let $\{f_n\}$ be nonnegative and measurable, and let $f_n \to f$ almost everywhere. Then

$$\int_E f \leq \liminf_{n \to \infty} \int_E f_n.$$

Corollary 4.1.
Let $\{f_n\}$ be nonnegative and measurable, with $f_n \to f$ almost everywhere and $0 \leq f_n \leq f$. Then $\int_E f = \lim_{n \to \infty} \int_E f_n$.

Theorem 4.6. *(Monotone Convergence)*
Let $\{f_n\}$ be nonnegative and measurable, and let f_n increase to f almost everywhere. Then f is measurable and

$$\int_E f = \lim_{n \to \infty} \int_E f_n.$$

Remark.
The theorem fails for decreasing sequences. If $f_n = \chi_{(n,+\infty)}$, then f_n decreases to 0, but $\int f_n = \infty$ while $\int f = 0$.

Corollary 4.2.
If $\{u_n\}$ are nonnegative and measurable and $\sum_{n=1}^{\infty} u_n = f$ almost everywhere, then

$$\int_E f = \sum_{n=1}^{\infty} \int_E u_n.$$

Corollary 4.3. *(Fatou)*
If $\{f_n\}$ are nonnegative and measurable, then

$$\int_E \liminf_{n\to\infty} f_n \leq \liminf_{n\to\infty} \int_E f_n.$$

Remark.
The functions $g_n = \inf_{k\geq n} f_k$ are nonnegative and measurable, increase to $\liminf f_n$, and satisfy $0 \leq g_n \leq f_n$. Monotone convergence and the comparison $\int g_n \leq \int f_n$ give the corollary.

Proposition 4.2.
A nonnegative measurable function that is integrable is finite almost everywhere.

Lemma 4.2. *(Beppo Levi)*
Let $\{f_n\}$ be nonnegative and measurable, and assume f_n is increasing. If $\left\{\int_E f_n\right\}$ is bounded, then f_n converges pointwise to an integrable function f, the limit is finite almost everywhere, and $\int_E f = \lim_{n\to\infty} \int_E f_n$.

4.3 Integrable Functions

Proposition 4.3.
Let f be measurable. Then $|f|$ is integrable if and only if both f^+ and f^- are integrable.

Proof.
Both f^+ and f^- are nonnegative and measurable, $0 \le f^{\pm} \le |f|$, and $|f| = f^+ + f^-$.
■

Definition 4.4. *(Integrable Function)*
A measurable function f is integrable if $|f|$ is integrable. In that case

$$\int_E f = \int_E f^+ - \int_E f^-.$$

Proposition 4.4.
An integrable function is finite almost everywhere. If f is integrable and $E_0 \subset E$ is a null set, then $\int_{E \setminus E_0} f = \int_E f$.

Proposition 4.5.
The integral of integrable functions is linear, additive over measurable subsets, monotone, and satisfies the triangle inequality.

Theorem 4.7. *(Dominated Convergence)*
Let $\{f_n\}$ be measurable, let g be integrable, and assume $|f_n| \le g$ almost everywhere. If $f_n \to f$ almost everywhere, then f is integrable and

$$\int_E f = \lim_{n \to \infty} \int_E f_n.$$

Theorem 4.8. *(Generalized Dominated Convergence)*
Let $f_n \to f$ almost everywhere, and let g_n be nonnegative and measurable with $g_n \to g$ almost everywhere. If $|f_n| \le g_n$ almost everywhere for every n and $\lim_{n\to\infty} \int_E g_n = \int_E g < \infty$, then f is integrable and $\int_E f = \lim_{n\to\infty} \int_E f_n$.

Theorem 4.9.
Let $\{f_k\}$ be integrable and $\sum_{k=1}^{\infty} \int_E |f_k| < \infty$. Then $\sum_k f_k(x)$ converges absolutely for almost every x, the sum is integrable, and

$$\int_E \sum_{k=1}^{\infty} f_k = \sum_{k=1}^{\infty} \int_E f_k.$$

Proof.
The function $F = \sum_k |f_k|$ is nonnegative and measurable, and monotone convergence gives $\int_E F = \sum_k \int_E |f_k| < \infty$. Thus F is finite almost everywhere, so the series of f_k converges absolutely almost everywhere. The partial sums are dominated by F, and dominated convergence passes the integral through the limit. ■

Theorem 4.10.
Let $E = \bigcup_{k=1}^{\infty} E_k$ with the E_k pairwise disjoint and measurable, and let f be integrable on E. Then

$$\int_E f = \sum_{k=1}^{\infty} \int_{E_k} f.$$

Corollary 4.4.
Let f be integrable on E, and let $\{E_k\}$ be measurable subsets of E. If $\{E_k\}$ increases to a set E_∞, then $\int_{E_\infty} f = \lim_k \int_{E_k} f$. If $\{E_k\}$ decreases to E_∞, then $\int_{E_\infty} f = \lim_k \int_{E_k} f$.

4.4 Absolute Continuity of the Integral

Lemma 4.3.
If $m(E) < \infty$, then for every $\delta > 0$ there is a finite partition of E into pairwise disjoint measurable sets of measure less than δ.

Proposition 4.6. *(Absolute Continuity of the Integral)*
Let f be measurable. If f is integrable, then for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$A \subset E \text{ measurable and } m(A) < \delta \quad \text{imply} \quad \int_A |f| < \varepsilon.$$

If $m(E) < \infty$, the converse holds.

Proof.
First assume f is integrable and $f \geq 0$. Given $\varepsilon > 0$, choose a bounded measurable f_ε with $0 \leq f_\varepsilon \leq f$, $m(\{f_\varepsilon \neq 0\}) < \infty$, and $0 \leq \int_E f - \int_E f_\varepsilon < \varepsilon/2$. Then $f_\varepsilon < M$ for some M. For $\delta = \varepsilon/(2M)$ and $m(A) < \delta$,

$$\int_A f = \int_A f_\varepsilon + \int_A (f - f_\varepsilon) < M m(A) + \frac{\varepsilon}{2} < \varepsilon.$$

Conversely, assume $m(E) < \infty$ and the ε-δ condition, and again take $f \geq 0$. Choose $\delta > 0$ so that $m(A) < \delta$ implies $\int_A f < 1$, and partition E into finitely many measurable pieces E_k of measure less than δ. Every bounded measurable h with $0 \leq h \leq f$ then satisfies

$$\int_E h = \sum_k \int_{E_k} h \leq \sum_k \int_{E_k} f < \infty.$$

The supremum defining $\int_E f$ is finite, so f is integrable. ■

Definition 4.5. *(Uniform Integrability)*
A family $\mathcal{F}$ of measurable functions is uniformly integrable if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$A \subset E, m(A) < \delta, f \in \mathcal{F} \quad \text{imply} \quad \int_A |f| < \varepsilon.$$

Example.
If g is nonnegative and integrable, then $\{f \text{ measurable} : |f| \leq g\}$ is uniformly integrable.

Theorem 4.11. *(Vitali Convergence)*
Let $m(E) < \infty$, and let $\{f_n\}$ be uniformly integrable. If $f_n \to f$ almost everywhere, then f is integrable and $\int_E f = \lim_{n\to\infty} \int_E f_n$.

Proposition 4.7.
Let $m(E) < \infty$, and let $\{h_n\}$ be nonnegative and measurable with $h_n \to 0$ almost everywhere. Then $\lim_{n\to\infty} \int_E h_n = 0$ if and only if $\{h_n\}$ is uniformly integrable.

Proposition 4.8.
If f is integrable, then for every $\varepsilon > 0$ there is a measurable $E_0 \subset E$ with $m(E_0) < \infty$ and $\int_{E\setminus E_0} |f| < \varepsilon$.

Definition 4.6. *(Tight Family)*
A family $\mathcal{F}$ of measurable functions is tight if for every $\varepsilon > 0$ there is a measurable $E_0 \subset E$ with $m(E_0) < \infty$ such that $\int_{E \setminus E_0} |f| < \varepsilon$ for every $f \in \mathcal{F}$.

Theorem 4.12. *(Vitali)*
Let $\{f_n\}$ be uniformly integrable and tight. If $f_n \to f$ almost everywhere, then f is integrable and $\int_E f = \lim_{n\to\infty} \int_E f_n$.

Corollary 4.5.
Let $\{h_n\}$ be nonnegative and measurable, and let $h_n \to 0$ almost everywhere. Then $\lim_n \int_E h_n = 0$ if and only if $\{h_n\}$ is uniformly integrable and tight.

4.5 Convergence in Measure

Definition 4.7. *(Convergence in Measure)*
Let $\{f_n\}$ be measurable and finite almost everywhere. The sequence converges to f in measure if, for every $\eta > 0$,

$$\lim_{n\to\infty} m(\{x \in E : |f_n(x) - f(x)| \geq \eta\}) = 0.$$

Proposition 4.9.
If f_n converges in measure to both f and g, then $f = g$ almost everywhere.

Proposition 4.10.
If f_n converges uniformly to f and f is finite almost everywhere, then f_n converges to f in measure.

Theorem 4.13. *(Lebesgue)*
Let $m(E) < \infty$. If $f_n \to f$ almost everywhere and f is finite almost everywhere, then f_n converges to f in measure.

Remark.
The converse is false. Enumerate the dyadic intervals

$$[0,1], \left[0,\frac{1}{2}\right], \left[\frac{1}{2},1\right], \left[0,\frac{1}{4}\right], \left[\frac{1}{4},\frac{1}{2}\right], \cdots$$

as $\{I_n\}$. Then χ_{I_n} converges to 0 in measure on $[0,1]$, but the sequence diverges at every point.

Theorem 4.14. *(Riesz)*
If f_n converges to f in measure, then some subsequence converges to f almost everywhere.

Proof.
For each $k \geq 1$, choose $n_k > n_{k-1}$ so that $m(E_k) < 2^{-k}$, where

$$E_k = \left\{ x \in E : |f(x) - f_{n_k}(x)| \geq \frac{1}{k} \right\}.$$

Then $\sum_k m(E_k) < \infty$. By Borel–Cantelli, almost every x lies in only finitely many E_k. At such an x there is $N(x)$ so that $k \geq N(x)$ implies $|f(x) - f_{n_k}(x)| \leq \frac{1}{k}$, and therefore $f_{n_k}(x) \to f(x)$. ■

Theorem 4.15.
Let $\{f_n\}$ be nonnegative and measurable. Then $\lim_{n\to\infty} \int_E f_n = 0$ if and only if f_n converges to 0 in measure and $\{f_n\}$ is uniformly integrable and tight.

Proof.
Suppose the integrals tend to 0, but some subsequence satisfies $\int_E f_{n_k} \geq \varepsilon$. That subsequence still converges to 0 in measure. Riesz's theorem extracts a further subsequence converging almost everywhere to 0, and Vitali's theorem forces its integrals to 0, a contradiction. Thus convergence in measure together with uniform integrability and tightness implies $\int_E f_n \to 0$.
Conversely, Chebyshev's inequality gives, for every $\eta > 0$,

$$m(\{f_n \geq \eta\}) \leq \left(\frac{1}{\eta}\right) \int_E f_n \to 0,$$

so $f_n \to 0$ in measure. For the tail of the sequence the integrals themselves are smaller than any prescribed ε, hence so are the integrals over every measurable subset and over the complement of every finite-measure subset. Each individual integrable function is absolutely continuous and tight, so the finite initial segment does not destroy uniform integrability or tightness. ■

Lemma 4.4.
Let $\{\varphi_n\}$ be integrable and increasing, and let $\{\psi_n\}$ be integrable and decreasing,

with $\varphi_n \le f \le \psi_n$ and $\lim_n \int_E (\psi_n - \varphi_n) = 0$. Then φ_n and ψ_n converge almost everywhere to f, the function f is integrable, and

$$\int_E f = \lim_n \int_E \varphi_n = \lim_n \int_E \psi_n.$$

Theorem 4.16.
Let $m(E) < \infty$ and let f be bounded. Then f is integrable if and only if f is measurable.

4.6 Riemann Integral

Definition 4.8. *(Oscillation)*
The oscillation of f at x is

$$\omega_f(x) = \inf_{\delta > 0} \sup\{|f(x_1) - f(x_2)| : |x_1 - x| < \delta, |x_2 - x| < \delta\}.$$

Remark.
- The function f is continuous at x if and only if $\omega_f(x) = 0$.
- The function ω_f is measurable.
- The set $C(f)$ of points at which f is continuous is a G_δ set.
- A set is the discontinuity set of some derivative if and only if it is an F_σ set of first category, that is, a countable union of nowhere dense closed sets.
- In the compact-open topology, the continuous functions that are differentiable at at least one point form a set of first category.

Lemma 4.5.
If f is bounded on an interval $I = [a, b]$, then

$$\int_I \omega_f = \overline{\int_a^b} f \, \mathrm{d}x - \underline{\int_a^b} f \, \mathrm{d}x.$$

Theorem 4.17. *(Lebesgue)*
A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if $m(\{x \in [a, b] : x \notin C(f)\}) = 0$.

5 Differentiation

5.1 Covering and Derivatives

Proposition 5.1.
A monotone function $f : (a, b) \to \mathbb{R}$ is continuous except at countably many points.

Proposition 5.2.
For every countable set $A \subset (a, b)$ there is a monotone function $f : (a, b) \to \mathbb{R}$ whose set of discontinuities is exactly A.

Proof.
Enumerate $A = \{r_n\}_{n=1}^{\infty}$ and set $f(x) = \sum_{r_n \le x} 2^{-n}$. ■

Proposition 5.3.
Every monotone function $f : (a, b) \to \mathbb{R}$ is measurable.

Proof.
Assume f is increasing. For each $r \in \mathbb{R}$ the set $A_r = \{x \in (a, b) : f(x) \le r\}$ is an interval: if $x_0, x_1 \in A_r$ and $x_0 \le x_1$, then every convex combination x_λ lies between them, so $f(x_\lambda) \le f(x_1) \le r$. An interval is measurable, so f is measurable. ■

Definition 5.1. *(Vitali Cover)*
Let $E \subset \mathbb{R}$. A family $\mathcal{F}$ of nondegenerate closed bounded intervals is a Vitali cover of E if for every $x \in E$ and every $\varepsilon > 0$ there is an interval $I \in \mathcal{F}$ with $x \in I$ and $l(I) < \varepsilon$.

Lemma 5.1. *(Vitali)*
Let $m^*(E) < \infty$, and let $\mathcal{F}$ be a Vitali cover of E. For every $\varepsilon > 0$ there exist finitely many pairwise disjoint intervals $I_1, \cdots, I_n \in \mathcal{F}$ such that

$$m^*\left(E \setminus \bigcup_{k=1}^{n} I_k\right) < \varepsilon.$$

Definition 5.2. *(Dini Derivates)*
The upper and lower derivates of f at x are

$$\overline{D}f(x) = \lim_{h\to 0}\sup_{0<|t|\le h}\frac{f(x+t)-f(x)}{t}, \quad \underline{D}f(x) = \lim_{h\to 0}\inf_{0<|t|\le h}\frac{f(x+t)-f(x)}{t}.$$

Remark.
Always $\underline{D}f(x) \le \overline{D}f(x)$. The two derivates agree if and only if f is differentiable at x.

Lemma 5.2.
Let $f:[a,b]\to\mathbb{R}$ be increasing. For every $\alpha>0$,

$$m^*\Big(\big\{x\in[a,b] : \overline{D}f(x)\ge\alpha\big\}\Big) \le \left(\frac{1}{\alpha}\right)(f(b)-f(a)).$$

Proof.
Write $E_\alpha = \{\overline{D}f \ge \alpha\}$. For each $\alpha' \in (0,\alpha)$, the closed intervals $[c,d]\subset[a,b]$ with $f(d)-f(c)\ge\alpha'(d-c)$ form a Vitali cover of E_α. Given $\varepsilon>0$, Vitali's lemma selects finitely many disjoint intervals $[c_k,d_k]$ from this cover with $m^*(E_\alpha\setminus\cup_k[c_k,d_k])<\varepsilon$. Then

$$\begin{aligned} m^*(E_\alpha) &\le \sum_k (d_k-c_k)+\varepsilon \\ &\le \left(\frac{1}{\alpha'}\right)\sum_k (f(d_k)-f(c_k))+\varepsilon \\ &\le \left(\frac{1}{\alpha'}\right)(f(b)-f(a))+\varepsilon. \end{aligned}$$

Let $\varepsilon\to 0$ and $\alpha'\to\alpha$. ■

Theorem 5.1. *(Lebesgue)*
An increasing function $f:[a,b]\to\mathbb{R}$ is differentiable almost everywhere.

Remark.
Riesz–Nagy. For every null set $E\subset[a,b]$ there is an increasing function $f:[a,b]\to\mathbb{R}$ that fails to be differentiable at every point of E.

Definition 5.3. *(Difference Quotient and Average)*
Let $f:[a,b]\to\mathbb{R}$ be integrable, and extend it by $f(x)=f(b)$ for $x\in(b,b+1]$. For $h\in(0,1]$ and $x\in[a,b]$,

$$\mathrm{Diff}_h f(x) = \frac{f(x+h)-f(x)}{h}, \quad A_h f(x) = \left(\frac{1}{h}\right)\int_x^{x+h} f.$$

Remark.
For every $[c,d] \subset [a,b]$, $\int_c^d \mathrm{Diff}_h f = A_h f(d) - A_h f(c)$.

Proposition 5.4.
Let $f : [a,b] \to \mathbb{R}$ be increasing, and assume f' is integrable on $[a,b]$. Then

$$\int_a^b f' \le f(b) - f(a).$$

Proof.
The difference quotients $\mathrm{Diff}_{\frac{1}{n}} f$ are nonnegative and measurable, and they converge almost everywhere to f'. Fatou's lemma gives

$$\int_a^b f' \le \liminf_{n\to\infty} n \int_a^b \left(f\left(x + \frac{1}{n}\right) - f(x) \right) \mathrm{d}x = f(b) - \limsup_{n\to\infty} n \int_a^{a+\frac{1}{n}} f.$$

Since f is increasing, the averages on $\left[a, a+\frac{1}{n}\right]$ are at least $f(a)$, and the right side is at most $f(b) - f(a)$. ■

Remark.
The inequality can be strict. The Cantor–Lebesgue function satisfies $\varphi' = 0$ almost everywhere, $\varphi(0) = 0$, and $\varphi(1) = 1$, so $\int_0^1 \varphi' < \varphi(1) - \varphi(0)$.

Proposition 5.5. *(Fubini)*
Let $f_n : [a,b] \to \mathbb{R}$ be increasing, and suppose $\sum_{n=1}^{\infty} f_n(x)$ converges for every $x \in [a,b]$. Then

$$\left(\sum_{n=1}^{\infty} f_n\right)'(x) = \sum_{n=1}^{\infty} {f_n}'(x)$$

for almost every $x \in [a,b]$.

5.2 Bounded Variation

Definition 5.4. *(Bounded Variation)*
For a partition $D : a = x_0 < \cdots < x_k = b$ of $[a, b]$, set $V_D(f) = \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})|$. The total variation is $V(f) = \sup V_D(f)$, the supremum running over all partitions. If $V(f) < \infty$, then f has bounded variation, written $f \in \mathrm{BV}[a, b]$.

Example.
- If f is increasing, then $V(f) = f(b) - f(a) < \infty$.
- If f is Lipschitz, then $V(f) \leq \lambda(b - a)$ for some $\lambda > 0$, so $\mathrm{Lip}[a, b] \subset \mathrm{BV}[a, b]$.
- The Cantor–Lebesgue function is not Lipschitz, but it is Hölder continuous: there exist $\lambda > 0$ and $\alpha \in (0, 1)$ such that $|\varphi(x) - \varphi(y)| \leq \lambda |x - y|^{\alpha}$.
- The function $f(x) = 2x \cos\left(\frac{\pi}{2x}\right)$ for $x \in (0, 1]$, with $f(0) = 0$, is continuous. On the partition with nodes $\frac{1}{2k}$, one has $f\left(\frac{1}{2k}\right) = \frac{(-1)^k}{k}$, and the variation of these nodes is a divergent harmonic series. Thus $f \notin \mathrm{BV}[0, 1]$.

Lemma 5.3.
For every $c \in (a, b)$, $V(f) = V(f; [a, c]) + V(f; [c, b])$.

Definition 5.5. *(Variation Function)*
If $f \in \mathrm{BV}[a, b]$, the variation function is $V_f(x) = V(f; [a, x])$.

Proposition 5.6.
If $f \in \mathrm{BV}[a, b]$, then both V_f and $f + V_f$ are increasing.

Proof.
The function V_f is increasing by the additivity of variation. If $a \leq x \leq y \leq b$, then

$$|f(y) - f(x)| \leq V(f; [x, y]) = V_f(y) - V_f(x),$$

so $f(y) + V_f(y) \geq f(x) + V_f(x)$. ■

Theorem 5.2. *(Jordan Decomposition)*
A function f lies in $\mathrm{BV}[a, b]$ if and only if $f = g - h$ for increasing functions g and h.

Proposition 5.7.
If $f \in \mathrm{BV}[a, b]$, then $\int_a^b |f'| \leq V(f)$.

5.3 Absolute Continuity

Definition 5.6. *(Absolute Continuity)*
A function $f : [a, b] \to \mathbb{R}$ is absolutely continuous, written $f \in \mathrm{AC}[a, b]$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$\sum_{k=1}^{n} |f(b_k) - f(a_k)| < \varepsilon$$

whenever $\{(a_k, b_k)\}_{k=1}^{n}$ are pairwise disjoint subintervals of (a, b) with $\sum_{k=1}^{n} (b_k - a_k) < \delta$.

Remark.
Every Lipschitz function is absolutely continuous, and every absolutely continuous function is continuous. On the compact interval $[a, b]$, continuity is equivalent to uniform continuity.

Example.
The function $f(x) = \sqrt{x}$ lies in $\mathrm{AC}[0, 1]$ but not in $\mathrm{Lip}[0, 1]$.

Proposition 5.8.
A function f lies in $\mathrm{AC}[a, b]$ if and only if V_f does.

Example.
The Cantor–Lebesgue function lies in $\mathrm{BV}[0, 1]$ but not in $\mathrm{AC}[0, 1]$.

Corollary 5.1.
A function f lies in $\mathrm{AC}[a, b]$ if and only if $f = g - h$ with g and h absolutely continuous and increasing.

Theorem 5.3.
Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f \in \mathrm{AC}[a, b]$ if and only if the family $\{\mathrm{Diff}_h f\}_{0<h<1}$ is uniformly integrable on $[a, b]$.

Theorem 5.4. *(Fundamental Theorem I)*
If $f \in \mathrm{AC}[a, b]$, then f' exists almost everywhere, f' is integrable, and

$$\int_a^b f' = f(b) - f(a).$$

Corollary 5.2. *(Integration by Parts)*
If $f, g \in \mathrm{AC}[a, b]$, then

$$\int_a^b fg' = f(b)g(b) - f(a)g(a) - \int_a^b gf'.$$

Theorem 5.5. *(Change of Variables)*
Let $u \in \mathrm{AC}[a, b]$ be increasing, and let f be integrable on $[u(a), u(b)]$. Then $(f \circ u)u'$ is integrable on $[a, b]$, and

$$\int_{u(a)}^{u(b)} f = \int_a^b (f \circ u)u'.$$

Proposition 5.9.
If $f \in \mathrm{AC}[a, b]$ and $E \subset [a, b]$ is a null set, then $f(E)$ is a null set.

Remark.
These conditions characterize absolute continuity together: $f \in \mathrm{AC}[a, b]$ if and only if f is continuous and of bounded variation and f sends null sets to null sets. An absolutely continuous function is differentiable almost everywhere and its derivative is integrable, but those two properties also hold for every function of bounded variation.

Definition 5.7. *(Indefinite Integral)*
A function $f : [a, b] \to \mathbb{R}$ is an indefinite integral if there is an integrable g on $[a, b]$ such that

$$f(x) = f(a) + \int_a^x g$$

for every $x \in [a, b]$.

Theorem 5.6.
A function $f : [a, b] \to \mathbb{R}$ is absolutely continuous if and only if it is an indefinite integral.

Corollary 5.3.
Let $f : [a, b] \to \mathbb{R}$ be monotone. Then $f \in \mathrm{AC}[a, b]$ if and only if $\int_a^b f' = f(b) - f(a)$.

Lemma 5.4.
Let $f : [a, b] \to \mathbb{R}$ be integrable. Then $f = 0$ almost everywhere if and only if $\int_c^d f = 0$ for every $[c, d] \subset [a, b]$.

Theorem 5.7. *(Fundamental Theorem II)*
Let $f : [a, b] \to \mathbb{R}$ be integrable and set $F(x) = \int_a^x f$. Then F is absolutely continuous, F is differentiable almost everywhere, and $F' = f$ almost everywhere.

Definition 5.8. *(Singular Function)*
A function $f \in \mathrm{BV}[a, b]$ is singular if $f' = 0$ almost everywhere.

Remark.
If $f \in \mathrm{AC}[a, b]$ is singular, then f is constant.

Theorem 5.8. *(Lebesgue Decomposition)*
Every $f \in \mathrm{BV}[a, b]$ can be written $f = g + h$ with $g \in \mathrm{AC}[a, b]$ and h singular.

Remark.
The singular part splits further as $h = u + v$, where u is continuous and singular and v is a jump function.

5.4 Convex Functions

Definition 5.9. *(Convex Function)*
A function $\varphi : (a, b) \to \mathbb{R}$ is convex if, for all $x_1, x_2 \in (a, b)$ and all $\lambda \in [0, 1]$,

$$\varphi(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda\varphi(x_1) + (1 - \lambda)\varphi(x_2).$$

Lemma 5.5.
Let $\varphi : [a,b] \to \mathbb{R}$ be convex, and let $a < x_1 < x_2 < b$. Writing $p = (x, \varphi(x))$ and $p_i = (x_i, \varphi(x_i))$, the slopes of the chords satisfy $S_{p,p_1} \le S_{p_1,p_2} \le S_{p,p_2}$.

Lemma 5.6.
If φ is convex, then $\varphi'_-(x)$ and $\varphi'_+(x)$ exist at every $x \in (a,b)$, and

$$\varphi'_-(x_1) \le \varphi'_+(x_1) \le \frac{\varphi(x_2)-\varphi(x_1)}{x_2-x_1} \le \varphi'_-(x_2) \le \varphi'_+(x_2)$$

whenever $x_1 < x_2$.

Corollary 5.4. *(Local Lipschitz)*
If $\varphi : (a,b) \to \mathbb{R}$ is convex, then φ is Lipschitz on every compact subinterval $[c,d] \subset (a,b)$.

Theorem 5.9.
A convex function $\varphi : (a,b) \to \mathbb{R}$ is differentiable except at countably many points, and φ' is increasing on the set where it exists.

Proposition 5.10.
Let ψ be increasing and integrable on $[a,b]$, and let $\alpha \in \mathbb{R}$. Then

$$\varphi(x) = \alpha + \int_a^x \psi$$

is convex.

Theorem 5.10. *(Jensen)*
Let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex, and suppose f and $\varphi \circ f$ are integrable on $[0,1]$. Then

$$\varphi\left(\int_0^1 f(x)\,\mathrm{d}x\right) \le \int_0^1 \varphi(f(x))\,\mathrm{d}x.$$

6 L^p Spaces

6.1 Norms

Definition 6.1. *(Norm)*
Let X be a linear space. A function $\|\cdot\| : X \to \mathbb{R}$ is a norm if
- $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$;
- $\|\alpha x\| = |\alpha| \, \|x\|$ for every scalar α;
- $\|x + y\| \leq \|x\| + \|y\|$.

Definition 6.2. *(L^p Space)*
Let E be measurable and $1 \leq p < \infty$. Then

$$L^p(E) = \left\{ f : E \to \mathbb{R} : \int_E |f|^p < \infty \right\},$$

with norm $\|f\|_p = \left(\int_E |f|^p\right)^{\frac{1}{p}}$. Functions that agree almost everywhere are identified.

Remark.
For $0 < p < 1$ the quantity $\|\cdot\|_p$ does not satisfy the triangle inequality. It does satisfy $\|f + g\|_p^p \leq \|f\|_p^p + \|g\|_p^p$. For $p \geq 1$ the space L^p is a Banach space, and for $p = 2$ it is a Hilbert space.

Definition 6.3. *(L^∞)*
For a measurable f,

$$\|f\|_\infty = \inf\{M \geq 0 : |f| \leq M \text{ a.e.}\},$$

and $L^\infty(E) = \{f : E \to \mathbb{R} : \|f\|_\infty < \infty\}$.

Definition 6.4. *(Sequence Space ℓ^p)*
For $1 \leq p < \infty$ and $a = \{a_k\}_{k=1}^\infty$ with $a_k \in \mathbb{R}$,

$$\ell^p = \left\{ a : \|a\|_p = \left(\sum_{k=1}^\infty |a_k|^p \right)^{\frac{1}{p}} < \infty \right\}.$$

For $p = \infty$, $\ell^\infty = \{a : \|a\|_\infty = \sup_k |a_k| < \infty\}$.

Definition 6.5. *(Space $C[a,b]$)*
On $C[a,b]$, the supremum norm is $\|f\|_\infty = \max_{x\in[a,b]}|f(x)|$. Convergence in this norm is uniform convergence.

6.2 Hölder and Minkowski

Definition 6.6. *(Conjugate Exponents)*
Exponents $p, q \in [1,\infty]$ are conjugate if $\frac{1}{p}+\frac{1}{q}=1$, with the convention $1/\infty = 0$.

Lemma 6.1. *(Young)*
Let $1 < p < \infty$ and let q be conjugate to p. If $a, b \geq 0$, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Theorem 6.1. *(Hölder)*
Let $1 \leq p \leq \infty$ and let q be conjugate to p. If $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and

$$\|fg\|_1 \leq \|f\|_p \ \|g\|_q.$$

Remark.
The same inequality holds in ℓ^p.

Corollary 6.1.
Let $1 \leq p < r < q < \infty$ and $f \in L^p \cap L^q$. Then $f \in L^r$, and

$$\log\|f\|_r \leq \lambda \log\|f\|_p + (1-\lambda)\log\|f\|_q,$$

where $\lambda = \frac{\frac{1}{r}-\frac{1}{q}}{\frac{1}{p}-\frac{1}{q}}$, or equivalently $\frac{1}{r} = \frac{\lambda}{p} + \frac{1-\lambda}{q}$.

Definition 6.7. *(Conjugate Function)*
Let $1 < p < \infty$, $f \in L^p$, and $f \neq 0$. The conjugate function of f is

$$f^* = \|f\|_p^{1-p} \ \operatorname{sgn}(f) \ |f|^{p-1}.$$

Lemma 6.2.
If p and q are conjugate and $f \in L^p$, then $f^* \in L^q$, $\|f^*\|_q = 1$, and $\int_E f f^* = \|f\|_p$.

Corollary 6.2. *(Minkowski)*
Let $1 \leq p \leq \infty$ and $f, g \in L^p$. Then $f + g \in L^p$ and $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Proposition 6.1.
Let $1 < p < \infty$ and let $\mathcal{F} \subset L^p$. If $\|f\|_p \leq M$ for every $f \in \mathcal{F}$, then $\mathcal{F}$ is uniformly integrable.

Proposition 6.2.
Let $m(E) < \infty$ and $1 \leq p_1 < p_2 \leq \infty$. Then $L^{p_2} \subset L^{p_1}$ and $\|f\|_{p_1} \leq c \, \|f\|_{p_2}$, where

$$c = \begin{cases} m(E)^{\frac{p_2 - p_1}{p_1 p_2}} & \text{if } p_2 < \infty \\ m(E)^{\frac{1}{p_1}} & \text{if } p_2 = \infty \end{cases}.$$

Remark.
- The function x^α is integrable on $(0, 1)$ if and only if $\alpha > -1$.
- The function x^α is integrable on $(1, +\infty)$ if and only if $\alpha \leftarrow 1$.
- If $1 \leq p_1 < p_2 \leq \infty$, then $\|a\|_\infty \leq \|a\|_{p_2} \leq \|a\|_{p_1} \leq \|a\|_1$, and therefore $\ell^1 \subset \ell^{p_1} \subset \ell^{p_2} \subset \ell^\infty$.

6.3 Completeness

Definition 6.8. *(Metric)*
Let X be nonempty. A function $\rho : X \times X \to \mathbb{R}$ is a metric if, for all $f, g, h \in X$,
- $\rho(f, g) \geq 0$, and $\rho(f, g) = 0$ if and only if $f = g$;
- $\rho(f, g) = \rho(g, f)$;
- $\rho(f, g) \leq \rho(f, h) + \rho(h, g)$.

Definition 6.9. *(Cauchy Sequence)*
A sequence $\{f_n\}$ in (X, ρ) is Cauchy if for every $\varepsilon > 0$ there is N such that $\rho(f_j, f_k) < \varepsilon$ whenever $j, k > N$.

Definition 6.10. *(Complete Metric Space)*
A metric space is complete if every Cauchy sequence converges.

Example.
A norm induces the metric $\rho(f,g) = \|f-g\|$. A complete normed space is a Banach space. The space $C[a,b]$ is complete under $\|\cdot\|_\infty$ and incomplete under $\|\cdot\|_1$.

Proposition 6.3.
A convergent sequence is Cauchy. A Cauchy sequence that has a convergent subsequence is itself convergent.

Definition 6.11. *(Rapidly Cauchy Sequence)*
A sequence $\{f_n\}$ is rapidly Cauchy if there exist $\varepsilon_n > 0$ with $\sum_{n=1}^{\infty} \varepsilon_n < \infty$ such that $\rho(f_n, f_{n+1}) < \varepsilon_n$ for every n.

Proposition 6.4.
Every Cauchy sequence has a rapidly Cauchy subsequence.

Theorem 6.2.
Let $1 \le p \le \infty$. If $\{f_n\}$ is rapidly Cauchy in L^p, then there exists $f \in L^p$ such that $f_n \to f$ almost everywhere and $\|f_n - f\|_p \to 0$.

Theorem 6.3. *(Riesz--Fischer)*
For $1 \le p \le \infty$ the space L^p is complete. Moreover, if $f_n \to f$ in L^p, then some subsequence converges to f almost everywhere.

Remark.
- Convergence in L^p implies convergence in measure when $1 \le p < \infty$, by Chebyshev's inequality. For $p = \infty$ the definition of the essential supremum yields a subsequence converging almost everywhere.
- Convergence in L^p does not imply almost-everywhere convergence. On $[0,1]$, the typewriter sequence

$$f_{2^k+j} = k\chi_{[j/2^k,(j+1)/2^k]}, \quad j = 0, \cdots, 2^k - 1,$$

converges to 0 in every L^p, but diverges at every x.

- Almost-everywhere convergence does not imply convergence in L^p. If $k \geq \frac{1}{p}$, the functions $f_n = n^k \chi_{(0,\frac{1}{n})}$ converge to 0 almost everywhere, while $\|f_n\|_p = n^{k-\frac{1}{p}}$ does not tend to 0.

Theorem 6.4.
Let $1 \leq p < \infty$ and $f_n, f \in L^p$, and assume $f_n \to f$ almost everywhere. Then $f_n \to f$ in L^p if and only if $\|f_n\|_p \to \|f\|_p$.

Theorem 6.5.
Let $1 \leq p < \infty$ and $f_n, f \in L^p$, and assume $f_n \to f$ almost everywhere. Then $f_n \to f$ in L^p if and only if $\{|f_n|^p\}$ is uniformly integrable and tight.

6.4 Dense Subclasses

Theorem 6.6.
Simple functions are dense in $L^p(E)$ for $1 \leq p < \infty$.

Theorem 6.7.
Step functions are dense in $L^p([a,b])$ for $1 \leq p < \infty$. A step function is a finite linear combination of characteristic functions of intervals.

Theorem 6.8.
Continuous functions of compact support are dense in $L^p([a,b])$ for $1 \leq p < \infty$.

Theorem 6.9.
Smooth functions of compact support are dense in $L^p([a,b])$ for $1 \leq p < \infty$.

Definition 6.12. *(Separable Space)*
A topological space is separable if it contains a countable dense subset.

Theorem 6.10.
If $E \subset \mathbb{R}$ is measurable and $1 \leq p < \infty$, then $L^p(E)$ is separable.

Remark.
The space $L^\infty([a,b])$ is not separable.

6.5 The Dual of L^p

Definition 6.13. *(Linear Functional)*
Let X be a linear space. A map $T : X \to \mathbb{R}$ is a linear functional if $T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$ for all $f, g \in X$ and all scalars α, β.

Definition 6.14. *(Bounded Linear Functional)*
Let $(X, \|\cdot\|)$ be a normed space. A linear functional $T : X \to \mathbb{R}$ is bounded if there exists $M \geq 0$ such that $|T(f)| \leq M\|f\|$ for every $f \in X$. Its operator norm is

$$\|T\| = \inf\{M \geq 0 : |T(f)| \leq M \ \|f\| \text{ for every } f \in X\}.$$

The space of all bounded linear functionals is the dual X^*, equipped with this norm.

Lemma 6.3.

$$\|T\| = \sup\{|T(f)| : f \in X, \|f\| \leq 1\} = \sup\{|T(f)| : f \in X, \|f\| = 1\}.$$

Corollary 6.3.
A linear functional is bounded if and only if it is continuous.

Proposition 6.5.
The dual $(X^*, \|\cdot\|)$ is a normed linear space.

Proposition 6.6.
Let $1 \leq p < \infty$, let q be conjugate to p, and let $g \in L^q$. The functional $T_g(f) = \int_E gf$ belongs to $(L^p)^*$ and $\|T_g\| = \|g\|_q$.

Proposition 6.7.
Let $T, S \in X^*$, and let $X_0 \subset X$ be dense. If $T(f) = S(f)$ for every $f \in X_0$, then $T = S$.

Lemma 6.4.
Let $1 \leq p < \infty$ and let q be conjugate to p. Suppose $g \in L^1$ and there is $M \geq 0$ such that $|\int_E gf| \leq M \|f\|_p$ for every simple $f \in L^p$. Then $g \in L^q$ and $\|g\|_q \leq M$.

Theorem 6.11.
Let $1 \leq p < \infty$ and let $T \in (L^p([a,b]))^*$. There exists $g \in L^q([a,b])$, with q conjugate to p, such that

$$T(f) = \int_a^b gf$$

for every $f \in L^p([a,b])$.

Remark.
The integrand is recovered from the indefinite integral $\varphi(x) = T\left(\chi_{[a,x]}\right)$. This φ is absolutely continuous, and $g = \varphi'$ lies in $L^q([a,b])$.

Remark.
If $A \subset E$ is measurable, then $\int_A f = \int_E f\chi_A$.

Theorem 6.12. *(Riesz Representation)*
Let $1 \leq p < \infty$, let q be conjugate to p, and let $E \subset \mathbb{R}$ be measurable. For every $T \in (L^p(E))^*$ there is a unique $g \in L^q(E)$ such that $T = T_g$ and $\|T\| = \|g\|_q$.